Roll No. | | | | | | | |

ODD END SEMESTER EXAMINATION DECEMBER – 2024

Name of the Course: M. TECH | Semester: I

Name of the Paper: CAD | Paper Code: MME 101

Time: 3 Hours | Maximum Marks: 100

Note:-

(i) All questions are compulsory.

(ii) Answer any two sub questions among a, b & c in each main question

(iii) Total marks in each main question are twenty.

(iv) Each question carries 10 marks

Q1	(20marks)	CO1
(a)	Discuss the importance of graphics standards in ensuring interoperability and quality in graphic systems?	
(b)	Differentiate between explicit, implicit, intrinsic, and parametric equations used in computer graphics with examples?	
(c)	Explain the various coordinate systems used in computer graphics and their applications in 2D and 3D modelling?	
Q2	(20 marks)	CO2
(a)	Differentiate between orthographic, oblique, and axonometric projections in 3D graphics with their applications?	
(b)	Rotate a point P (3,4,5) about an axis passing through the origin and aligned with the direction vector V(1,1,1) by 90^0?	
(c)	A point P (5,4,10) is projected onto a perspective plane z=2 with the projection reference point at the origin. Calculate the coordinates of the projected point?	
Q3	(20 marks)	CO3
(a)	Compare Bezier and B-Spline curves in terms of their generation techniques, control points, and properties? Highlight their applications in CAD modelling?	
(b)	What are rational curves and NURBS? Explain their mathematical formulation and importance in representing freeform shapes?	
(c)	For a quadratic B-Spline with control points P0 (1,1), P1 (2,2), P2 (3,1), P3 (4,0) and a uniform knot vector U=[0,0,0,1,2,3,3,3], compute the curve point at u=1.5?	
Q4	(20 marks)	CO4 & CO5
(a)	Explain the construction and applications of plane surfaces, cylindrical surfaces, and ruled surfaces in CAD systems?	
(b)	Generate a cylindrical surface using a circle of radius r=2, centered at (0,0), and extruded along the z-axis from z=0 to z=5. Represent the parametric equation of the surface and calculate the coordinates for u=π/4 and v=3 ?	
(c)	Discuss the concept of continuity in solid modelling? How are composite solids formed in CAD systems?	
Q5	(20 marks)	CO5 & CO6
(a)	Compare wireframe, surface, and solid modeling systems in terms of their features, applications, and limitations?	
(b)	What is the parametric representation of solids? Explain its advantages and provide an example?	
(c)	What are the features of web-based modeling? How does simplification of assemblies benefit CAD applications?	